Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

Addressing Key Business Challenges

How can we leverage consumer shopping data to optimize our strategies?

A leading retail company aims to deepen its understanding of customer behavior to enhance sales, satisfaction, and loyalty. Management has observed shifts in purchasing patterns across various demographics, product categories, and sales channels.

We focused on identifying key drivers like discounts, reviews, seasonality, and payment preferences that influence consumer decisions and repeat purchases.

Project Deliverables

Data Preparation & Modeling

Clean, transform, and load raw data into a structured database using Python.

Data Analysis (SQL)

Extract insights on customer segments, loyalty, and purchase drivers through 10 targeted SQL queries.

Visualization & Insights

Develop an interactive Power BI dashboard to highlight key patterns and present actionable recommendations.

Project Overview & Data Snapshot

This project analyzes customer shopping behavior using transactional data to identify trends and inform strategic decisions.

3,900

Total Customers

Analyzed transactions

$59.7

Avg Purchase

Average transaction value

3.75

Avg Rating

Customer satisfaction score

4

Product Categories

Analyzed across the dataset

The dataset comprises 3,900 rows and 18 columns, detailing demographics, purchase specifics, and behavioral attributes.

Data Preparation & Engineering Workflow

Load Data

Missing Values

Standardize Columns

Remove Duplicates

Age Group Feature

Our robust data preparation process ensures data quality and analytical readiness. Key steps included handling missing values, standardizing column names, removing redundancies, and engineering new features like **Age Group** and **Purchase Frequency Days** for deeper insights.

Quantile-based Age Binning

`pd.qcut()` ensured balanced demographic segments.

Group-wise Median Imputation

Preserved category-level review rating distributions.

Redundancy Elimination

Dropped duplicate `promo_code_used` column to avoid noise.

SQL Analysis: Key Findings

SQL INSIGHTS

Revenue by Gender

Male customers generate **2.1x higher revenue** ($157,890) compared to female customers ($75,191). This suggests a need for targeted female-centric marketing.

Gender	Revenue (USD)
Female	$75,191
Male	$157,890

High-Spending Discount Users

We identified 839 customers who used a discount and spent above the average purchase amount. This segment represents a valuable opportunity for strategic discount optimization.

Top 5 Products by Review Rating

Gloves, Sandals, Boots, Hat, and Skirt are our top-rated products, indicating strong customer satisfaction and potential for premium positioning.

Rank	Product	Avg Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Shipping Type Comparison

Express shipping correlates with a slightly higher average purchase amount ($60.48) compared to Standard ($58.46).

Shipping Type	Avg Purchase ($)
Standard	$58.46
Express	$60.48

Deep Dive: Discounts & Subscriptions

SQL INSIGHTS

Subscribers vs. Non-Subscribers

While subscribers (27% of customers) generate substantial revenue, their average spend is nearly identical to non-subscribers. This suggests the subscription program needs re-evaluation.

Subscription Status	Total Customers	Avg Spend ($)	Total Revenue ($)
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Discount-Dependent Products

Products like Hats, Sneakers, and Coats show high discount dependency, with nearly half of their sales relying on promotions. This indicates potential margin erosion and calls for a review of pricing strategies.

Rank	Product	Discount Rate (%)
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

Customer Segmentation & Product Popularity

SQL INSIGHTS

Customer Segmentation

A remarkable **79.9% of our customer base is 'Loyal'** (over 10 purchases), indicating a strong retention foundation. However, only 2.1% are 'New,' highlighting a need to boost acquisition efforts.

Segment	Customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Repeat Buyers & Subscriptions

Among customers with >5 purchases, 2,518 are not subscribed, presenting a clear opportunity to convert them to our subscription program.

Subscription Status	Repeat Buyers (>5 purchases)
No	2,518
Yes	958

Top Products per Category

Accessories: Jewelry, Sunglasses, Belt
Clothing: Blouse, Pants, Shirt
Footwear: Sandals, Shoes, Sneakers
Outerwear: Jacket, Coat, Hoodie

Jewelry and Blouse lead with 171 orders each. This granular view helps optimize inventory and merchandising for each category.

Revenue by Age Group: Balanced Contribution

SQL INSIGHTS

Revenue contribution is relatively balanced across all age groups, ranging from $55K to $62K, indicating broad appeal for our product offerings.

Young Adults lead with $62,143, closely followed by **Middle-aged** customers at $59,197. These segments should be prioritized for targeted campaigns due to their higher spending.

Rank	Age Group	Total Revenue ($)
1	Young Adult	$62,143
2	Middle-aged	$59,197
3	Adult	$55,978
4	Senior	$55,763

Power BI Dashboard: Visualizing Insights

Our interactive Power BI dashboard provides a dynamic view of customer behavior. Slicers for Subscription Status, Gender, Category, and Shipping Type allow for real-time cross-filtering and deep-dive analysis across all charts.

KPI Cards

3,900 Customers, $59.76 Avg Purchase, 3.75 Avg Rating.

Subscription Donut

73% Non-subscribers vs 27% Subscribers.

Revenue by Category

Clothing leads with over $100K in revenue.

Revenue by Age Group

Young Adults top revenue, all groups clustered $55K-$62K.

Key Insights & Recommendations

Boost Subscription Program

Target repeat buyers with a free trial and promote exclusive benefits to convert the 72% of repeat buyers who are not subscribed.

Review Discount Policy

Shift from always-on promotions to time-limited flash discounts, especially for products like Hats and Sneakers, to preserve margins and perceived value.

Shipping Upsell Opportunity

Implement a checkout nudge for Express shipping, highlighting its benefits for a slight increase in average transaction value.

Activate Returning Customers

Deploy personalized email sequences and "Complete your loyalty journey" messages to convert the 701 returning customers into loyal advocates.

Spotlight Top-Rated Products

Feature Gloves, Sandals, and Boots prominently in marketing campaigns and product listing pages to leverage social proof and boost sales.

Age-Targeted Marketing

Personalize catalog views and marketing messages for Young Adults and Middle-aged segments, aligning product recommendations with their preferences.